Dr. XIAN, Poline Yin

冼殷博士

BEng, MSc, PhD
Lecturer, Department of Computer Science

polinexian@comp.hkbu.edu.hk

RRS 717 +852 3411 5928

About

Dr. Xian obtained her Ph.D. and M.Sc. degrees in Electrical and Computer Engineering from Duke University. Before joining Hong Kong Baptist University, she was a research scientist at the Hong Kong Science Park. She has also been a lecturer at Hong Kong University of Science and Technology (HKUST) and Duke University, and a research scholar at the Hausdorff Institute of Mathematics in Germany. Dr. Xian's developed products have received the Hong Kong ICT awards, and secured funding from Lenovo, Sun Hung Kai Properties, and Hong Kong ITF. She generated patents and technical standard reports in numerical optimization and wireless sensing. Dr. Xian served as a technical committee member and reviewer in international conferences and journals, including the International Joint Conference on Artificial Intelligence (IJCAI), and the International Conference on Learning Representation (ICLR). She is recognized as Fellow of Higher Education Academy (FHEA) in 2025.

Research Interests

- Audio, image, and signal processing
- High-dimensional data analysis
- Non-convex optimization and application

Selected Publications

- **Y. Xian**, H. Liu, X. Tai, and Y. Wang. "Randomized Kaczmarz Method for Single Particle X-Ray Image Phase Retrieval." *Handbook of Mathematical Models and Algorithms in Computer Vision and Imaging: Mathematical Imaging and Vision*. Cham: Springer International Publishing, 2023.
- H. Gu, **Y. Xian**, I. C. Unarta, and Y. Yao. "Generative adversarial networks for robust cryo-EM image denoising." *Handbook of mathematical models and algorithms in computer vision and imaging: mathematical imaging and vision*. Cham: Springer International Publishing, 2023
- **Y. Xian**, H. Gu, W. Wang, X. Huang, Y. Yao, Y. Wang, and J.F. Cai. "Data-driven tight frame for Cryo-EM image denoising and conformational classification." *IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, 2018.
- **Y. Xian**, Y. Pu, Z. Gan, L. Lu, and A. Thompson, "Adaptive DCTNet for audio signal classification." *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2017.
- **Y. Xian**, A. Thompson, Q. Qiu, L. Nolte, D. Nowacek, J. Lu, and R. Calderbank, "Classification of whale vocalizations using the Weyl transform." *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2015.